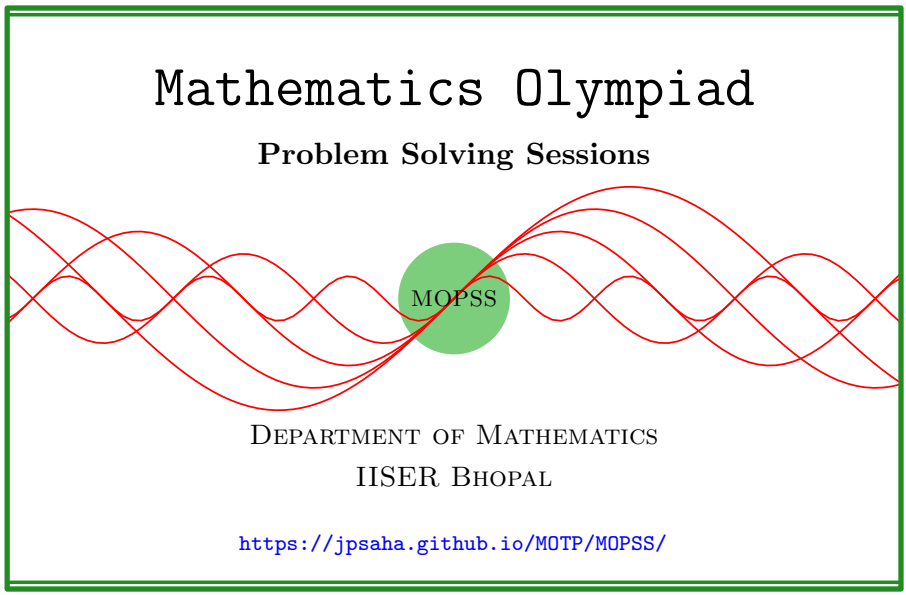
Coloring proofs

MOPSS



Suggested readings

- Evan Chen's advice On reading solutions, available at https://blog.evanchen.cc/2017/03/06/on-reading-solutions/.
- Evan Chen's Advice for writing proofs/Remarks on English, available at https://web.evanchen.cc/handouts/english/english.pdf.
- Notes on proofs by Evan Chen from OTIS Excerpts [**Che25**, Chapter 1].
- Tips for writing up solutions by Edward Barbeau, available at https://www.math.utoronto.ca/barbeau/writingup.pdf.
- Evan Chen discusses why math olympiads are a valuable experience for high schoolers in the post on Lessons from math olympiads, available at https://blog.evanchen.cc/2018/01/05/lessons-from-math-olympiads/.

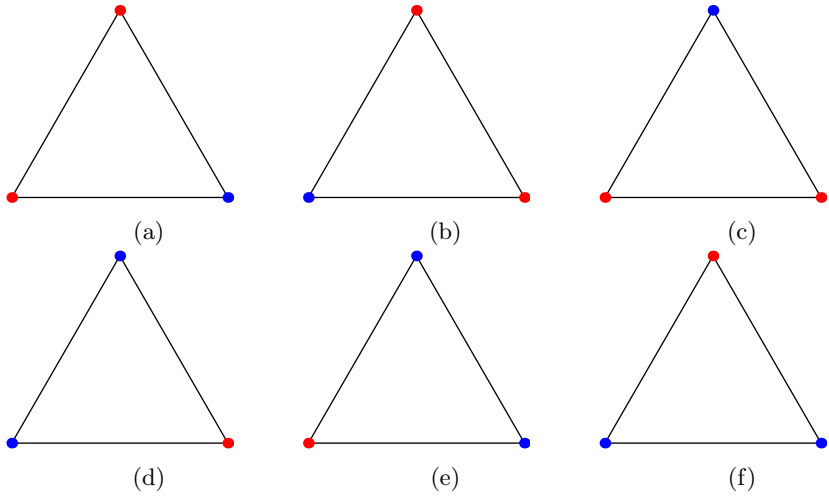


Figure 1: Two monochromatic points are one unit apart, Example 1

List of problems and examples

3 Exercise (Bay Area MO 2007 P2) 4

§1 Coloring proofs

See [**Sob13**, §3.2], [**Eng98**, Chapter 2].

§1.1 Two coloring

Example 1. If each point of the plane is colored red or blue, then there are two points of the same color at distance 1 from each other.

Summary — Consider a "bigger" or an "auxiliary" structure such that its size puts some constraint on it (for instance, by the pigeonhole principle), which would lead to the result. See Fig. 1.

Solution 1. Consider an equilateral triangle with side-length 1. Then at least two of its vertices are of the same color, which proves the result. ■

Example 2. Suppose that to every point of the plane a colour, either red or blue, is associated.

1. Show that if there is no equilateral triangle with all vertices of the same colour then there must exist three points A, B and C of the same colour such that B is the mid-point of AC.

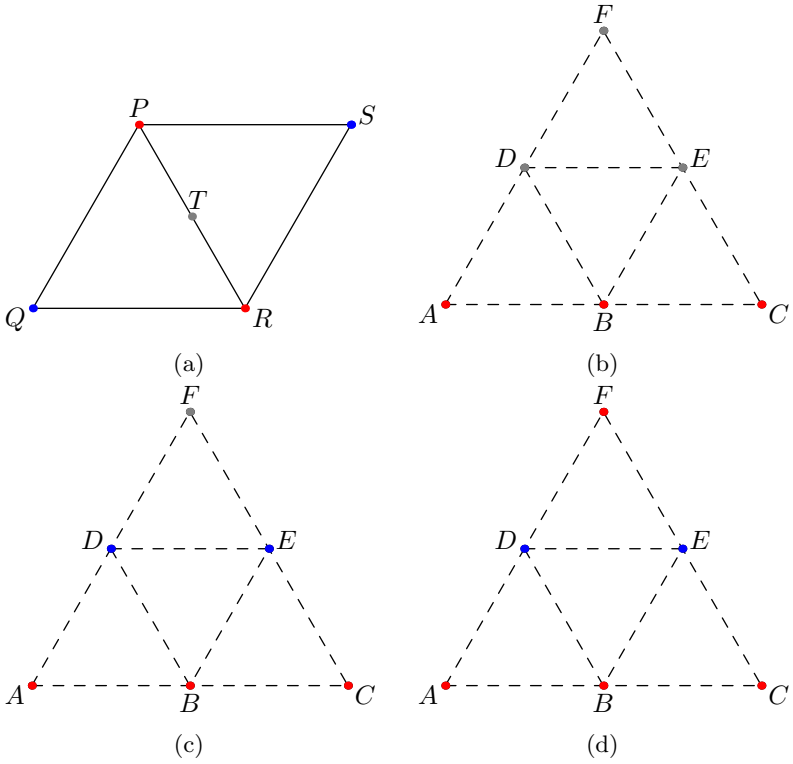


Figure 2: Example 2

2. Use the above to conclude that there must be an equilateral triangle with all vertices of the same colour.

Summary — Consider a "bigger" or an "auxiliary" structure such that its size puts some constraint on it (for instance, by the pigeonhole principle), which would lead to the result. See Fig. 2.

Solution 2. Let us prove the first part. Suppose there is no equilateral triangle with all vertices of the same color. So any equilateral triangle has two vertices of the same color. Let PQR be an equilateral triangle such that P, R are of the same color. Without loss of generality[1], assume that P, R are red. Note that Q is blue. Consider the equilateral triangle PRS as in Fig. 2a. Then S is blue. Let T denote the mid-point of PR. If T is red (resp. blue), then we could take (A, B, C) equal to (P, T, R) (resp. (Q, T, S)).

Suppose there is no equilateral triangle with all vertices of the same color. By the first part, there exist three points A, B, C such that B is the mid-point

[1]Is it clear that there is *no loss of generality*?

of AC (as in Fig. 2b). Let F be a point on the plane such that ACF is an equilateral triangle. Let D (resp. E) denote the mid-point of AF (resp. CF). Since ABD is equilateral, D is blue (as in Fig. 2c). Similarly, E is also blue (as in Fig. 2c). Hence F is red (as in Fig. 2d). So the triangle ACF has the required properties. Therefore, assuming that there is no equilateral triangle with all vertices of the same color, we proved that there is an equilateral triangle with all vertices of the same color. This shows that there is an equilateral triangle with all vertices of the same color. ■

Exercise 3 (Bay Area MO 2007 P2)**.** The points of the plane are colored in red and blue so that whenever three vertices of a parallelogram are the same color, the fourth vertex is that color, too. Prove that all the points of the plane are the same color.

Solution 3. If not, then choose two points A, B from the plane which are of different color as in Fig. 3. Let C denote the mid-point of AB. Without loss of generality, we assume that B, C are of the same color, say red. Then A is blue. Draw a parallelogram $BCDE$. Note that if E were red, then considering the parallelogram $BCDE$, it would follow that D is red, and then considering the parallelogram $ACED$, we would obtain that A is red. Hence, E is blue. Since BC, DE are parallel and have the same length, so are AC and DE, and hence $ACDE$ is a parallelogram. Considering the parallelogram $ACDE$ once again, it follows that D is blue. Since A, D, E are blue, C cannot be red. Hence all points of the plane are of the same color. ■

§1.2 Three coloring

Example 4. Suppose that each point in the plane is colored red, green or blue. Prove that either there are two points of the same color a distance 1 unit apart, or there is an equilateral triangle of side length $\sqrt{3}$ all of whose vertices are of the same color.

> **Summary** — Consider a "bigger" or an "auxiliary" structure such that its size puts some constraint on it (for instance, by the pigeonhole principle), which would lead to the result. See Figs. 4 and 5.

Solution 4. Note that it suffices to prove the claim below.

> **Claim** — Suppose that each point in the plane is colored red, green or blue. If no two points, which are one unit apart, are of the same color, then there is an equilateral triangle of side length $\sqrt{3}$ all of whose vertices are of the same color.

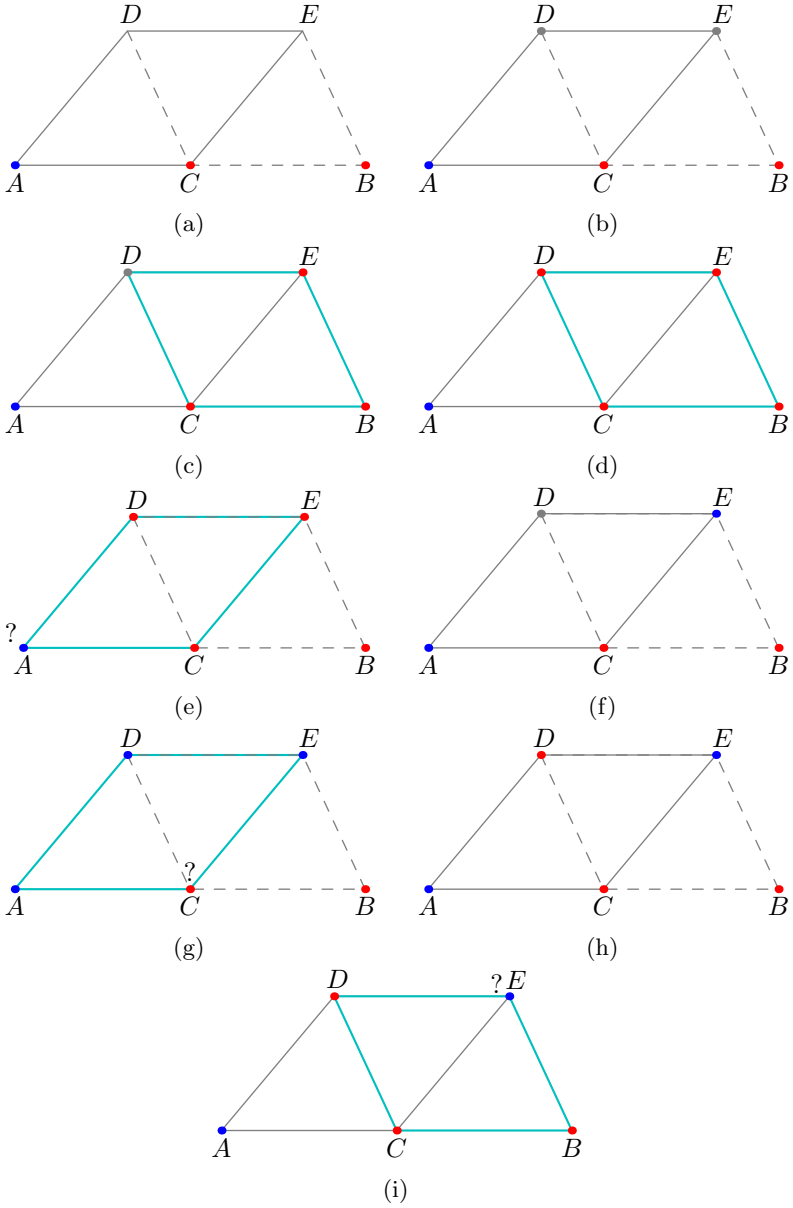


Figure 3: Bay Area MO 2007 P2, Exercise 3

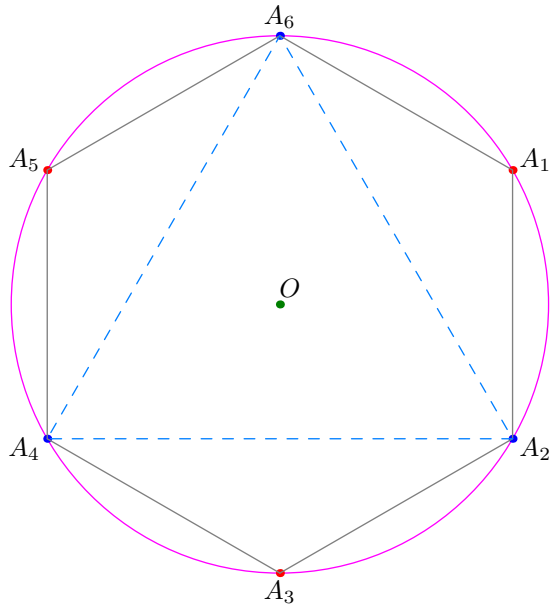


Figure 4: Example 4

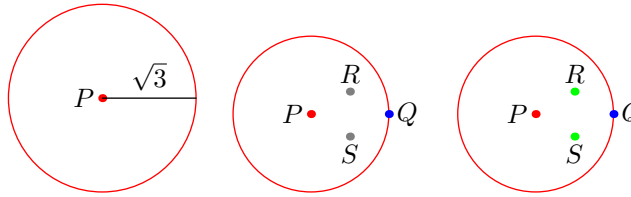


Figure 5: ?? 1

Proof of the Claim. Assume that no two points which are one unit apart are of the same color. Choose a point O from the plane and without loss of generality assume that O is green. Then draw a unit circle with centre at O, and a regular hexagon circumscribed in this unitcircle as shown in Fig. 4. By the assumption, none of its vertices are green, and hence a vertex is red or blue. Since the edges of the hexagon are of length one, it follows that the colors of its vertices are red and green alternatively. Then the triangle $A_2A_4A_6$ is an equilateral triangle of side length $\sqrt{3}$ and all of its vertices are of the same color. This proves the claim. □

■

Remark 1. In fact, one can show that when the points of the plane are three colored, then there are two monochromatic points that are one unit apart.

> **Claim** — Suppose that each point in the plane is colored red, green or blue. Then there are two monochromatic points that are one unit apart.

Proof of the Claim. On the contrary, let us assume that no two points, which are one unit apart, are monochromatic. Let us consider a point P on the plane. Suppose it is red, and consider the circle of radius $\sqrt{3}$ with centre at P.

In fact, assuming each point in the plane is colored red, green or blue, one can show that there are two monochromatic points that are one unit apart. Otherwise, we choose a point P, suppose it is red, and consider the circle $\mathcal{C}$ of radius $\sqrt{3}$ with centre at P (as in Fig. 5a).

> **Subclaim** — All points on the circumference of $\mathcal{C}$ are red.

Proof of the SubClaim. On the contrary, let us suppose that there is a point on the circumference of $\mathcal{C}$ which is red. Consider one such point Q. Suppose it is blue. Consider the points R, S (as in Fig. 5b), which lie on the opposite sides of PQ such that RS is perpendicular to PQ and PRS is an equilateral triangle. Note that the point R has to be green. Indeed, if R is red (resp. green), then the points P, R (resp. Q, R) are monochromatic and one unit apart, which contradicts our assumption. Similarly, the point S also has to be green (as in Fig. 5c). Note that R, S are one unit apart. Since they are monochromatic, we obtain a contradiction to our assumption. This proves the subclaim. □

Consider any chord of $\mathcal{C}$ of length one unit. The end-points of the chord are red by the SubClaim. The claim follows. □

References

[**Che25**] Evan Chen. *The OTIS Excerpts*. Available at https://web.evanchen.cc/excerpts.html. 2025, pp. vi+289 (cited p. 1)

[**Eng98**] Arthur Engel. *Problem-solving strategies*. Problem Books in Mathematics. Springer-Verlag, New York, 1998, pp. x+403. ISBN: 0-387-98219-1 (cited p. 2)

[**Sob13**] Pablo Soberón. *Problem-solving methods in combinatorics*. An approach to olympiad problems. Birkhäuser/Springer Basel AG, Basel, 2013, pp. x+174. ISBN: 978-3-0348-0596-4; 978-3-0348-0597-1. DOI: 10.1007/978-3-0348-0597-1. URL: http://dx.doi.org/10.1007/978-3-0348-0597-1 (cited p. 2)